

AMPHIBIANS

The Age of Amphibians—when amphibian giants ruled the planet—belongs to the prehistoric past. However, amphibians still include more than 7,000 species today—a thousand more than all the known mammals. Moist, glandular skin helps many amphibians to absorb oxygen and protects some with deadly

poison. It also makes them vulnerable to dehydration and prevents them from entering salty ocean waters. However, in moist habitats such as rainforests, amphibians have evolved into a multitude of frogs, toads, and salamanders, which includes some of the most exquisitely colorful species on the planet.

Anatomy

Six out of seven kinds of amphibians are frogs or toads, and they embody the essential aspects of amphibian biology: adults have lungs but can also breathe through their moist skin. Their soft eggs usually hatch into aquatic larvae. The frog's build—with its long hindlegs—is adapted for jumping, but many species use their legs for digging or even swimming.

Other groups of living amphibians are the wormlike caecilians and lizardlike newts and salamanders. Burrowing caecilians lack limbs, and their tiny eyes are sometimes hidden by a heavily roofed skull, which is used for bulldozing through soil. Salamanders—and their more aquatic relatives, the newts—are generally crawling amphibians with long tails. A few lack hindlegs and are eel-like swimmers.

As amphibians lack the sharp teeth and claws of reptiles, many rely on chemical defense to deter predators. Their skin is peppered with numerous glands, many of which secrete slime, but others produce toxins. The so-called “poison frogs” are notorious for their deadly toxicity—and advertise the fact by flashing bright warning colors.

▽ COLORFUL WARNING

Otherwise largely defenseless, amphibians rely on poisonous skin to deter predators. Brightly colored poison frogs accumulate their potent toxins from the invertebrates they eat.

large, black eyes



△ ATTRACTING A MATE

Courtship in amphibians can be elaborate. Male frogs call for attention, but the smooth newt-like related species—uses color to impress a mate.

Reproduction

Amphibian eggs are coated with soft jelly that is not drought-resistant. This means that the eggs must be laid in water or moist places. Almost all frogs and toads—and most primitive salamanders—reproduce by external fertilization: sperm and eggs are released into water. Males of more advanced salamanders and newts deposit a packet of sperm, which the female manipulates into her genital opening. Caecilians, by contrast, are some of the only amphibians that practice internal fertilization by direct copulation.

Many rainforest frogs breed in puddles or pools that collect in leaves. Others lay eggs on moist ground—or in elaborate bubble nests made in vegetation, relying on their tadpoles being able to drop into a pool of water underneath. Some kinds hatch into tiny froglets instead of tadpoles and a few parents even carry their eggs or offspring around with them, piggyback fashion. A range of developmental changes occurs in salamanders: the eggs directly hatch into miniature versions of the parents, while a few give birth to live young.



◁ FIRE BELLY

Some amphibians use shock tactics when under threat. The fire-bellied toad flashes its orange underside using various gymnastics—including flipping onto its back.

Behavior

The aquatic larvae of amphibians retain some of the sensory features of fish—including their vibration-detecting lateral line system, which senses water currents. But the sensory emphasis in land-living adults is different. Burrowing caecilians rely on a sense of smell to hunt underground invertebrates or find mates—and have a tiny tentacle below each eye to collect this information. Frogs and toads usually have good vision and a very sharp sense of hearing—evident from their big eyes and conspicuous eardrums. Many are highly vocal and amplify their calls with bulbous, air-filled vocal sacs. Calls—ranging from insectlike chirps to barks and metallic peals—are species specific and the principal means of communication between individuals, especially when breeding.

Salamanders and newts rely more on underwater pheromones or sometimes visual cues. Their elaborate courtship rituals often involve flaunting bright, dramatic colors.

▷ SPAWNING SITE

Many amphibians release sperm and eggs in open water. Communally breeding frogs spawn in thick masses, warming the eggs to speed up hatching.

AMPHIBIAN CLASSIFICATION

There are three orders of amphibians: frogs and toads, the largest group; the lizardlike newts and salamanders; and the wormlike caecilians.

Newts and salamanders » p438
Order Caudata Species 691

Caecilians » p439
Order Gymnophiona Species c.203

Frogs and toads » pp439–41
Order Anura Species 6,481